Exercises: Phenomena in High Dimensions – Answer Key

Mathematical Foundations for Machine Learning

Volume of ℓ_p-Balls and the Gamma Function

Exercise 1.1 [Easy]

The volume of the unit ℓ_p-ball in $\mathbb{R}^d$ is given by

$$V_d^p(1) = \frac{\left(2\,\Gamma\left(\frac{1}{p}+1\right)\right)^d}{\Gamma\left(\frac{d}{p}+1\right)}.$$

(a) Using the identity $\Gamma(x+1) = x\,\Gamma(x)$ and $\Gamma(1) = 1$, compute $\Gamma(4)$.
(b) Using $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, compute $\Gamma\left(\frac{5}{2}\right)$.
(c) Compute the exact volume of the unit ℓ_2-ball in $\mathbb{R}^4$.

Solution

(a) Using the recurrence $\Gamma(x+1) = x\,\Gamma(x)$:

$$\Gamma(2) = 1 \cdot \Gamma(1) = 1, \quad \Gamma(3) = 2 \cdot \Gamma(2) = 2, \quad \Gamma(4) = 3 \cdot \Gamma(3) = 6.$$

Indeed, $\Gamma(n+1) = n!$ so $\Gamma(4) = 3! = 6$.
(b) Starting from $\Gamma(\frac{1}{2}) = \sqrt{\pi}$:

$$\Gamma\left(\tfrac{3}{2}\right) = \tfrac{1}{2}\,\Gamma\left(\tfrac{1}{2}\right) = \frac{\sqrt{\pi}}{2}, \qquad \Gamma\left(\tfrac{5}{2}\right) = \tfrac{3}{2}\,\Gamma\left(\tfrac{3}{2}\right) = \frac{3\sqrt{\pi}}{4}.$$

(c) With $p = 2$ and $d = 4$:

$$V_4^2(1) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} = \frac{\pi^2}{\Gamma(3)} = \frac{\pi^2}{2}.$$

Alternatively, using the general formula with $\Gamma(3/2) = \sqrt{\pi}/2$: $V_4^2(1) = (2 \cdot \sqrt{\pi}/2)^4/\Gamma(3) = \pi^2/2$.

Exercise 1.2 [Easy]

Recall that the volume of the unit ℓ_1-ball in $\mathbb{R}^d$ is $V_d^1(1) = \frac{2^d}{d!}$.
(a) Compute $V_d^1(1)$ for $d = 2, 4, 6, 10$ and describe the qualitative behavior.
(b) The volume of the unit ℓ_∞-ball in $\mathbb{R}^d$ is $V_d^\infty(1) = 2^d$. Compare the growth behaviors of $V_d^1(1)$, $V_d^2(1)$, and $V_d^\infty(1)$ as $d \to \infty$. Which of these converge to zero?

Solution

(a)

$$V_2^1(1) = \frac{4}{2} = 2, \quad V_4^1(1) = \frac{16}{24} = \frac{2}{3} \approx 0.67, \quad V_6^1(1) = \frac{64}{720} \approx 0.089, \quad V_{10}^1(1) = \frac{1024}{3628800} \approx 2.8\times10^{-4}.$$

The volume decreases rapidly toward zero as d increases. The factorial in the denominator grows much faster than the exponential 2^d in the numerator.
(b) As $d \to \infty$:

- $V_d^1(1) = 2^d/d! \to 0$ (factorial dominates exponential),
- $V_d^2(1) = \pi^{d/2}/\Gamma(d/2+1) \to 0$ (same factorial-type decay via Gamma function),
- $V_d^\infty(1) = 2^d \to \infty$ (exponential growth with no denominator suppression).

So both ℓ_1 and ℓ_2 unit ball volumes vanish, while the ℓ_∞ unit ball volume grows without bound. Moreover, $V_d^1(1)$ goes to zero faster than $V_d^2(1)$.

Exercise 1.3 [Medium]

The volume of the ℓ_p-ball of radius r in $\mathbb{R}^d$ scales as $V_d^p(r) = V_d^p(1)\, r^d$.

(a) Show that the radius r_d such that $V_d^1(r_d) = 1$ satisfies

$$r_d = \frac{(d!)^{1/d}}{2}.$$

(b) Using Stirling's approximation $n! \approx \sqrt{2\pi n}\,(n/e)^n$, show that for large d,

$$r_d \approx \frac{d}{2e}.$$

(c) Similarly, for the ℓ_2-ball, one can show that the unit-volume radius scales as $r_d \approx \sqrt{d/e}$ for even dimensions. Interpret why $r_d \to \infty$ for both ℓ_1 and ℓ_2, and explain why this phenomenon does *not* occur for the ℓ_∞-ball.

Solution

(a) We need $V_d^1(r_d) = V_d^1(1)\, r_d^d = \frac{2^d}{d!}\, r_d^d = 1$. Solving:

$$r_d^d = \frac{d!}{2^d} \implies r_d = \left(\frac{d!}{2^d}\right)^{1/d} = \frac{(d!)^{1/d}}{2}.$$

(b) Applying Stirling's approximation $d! \approx \sqrt{2\pi d}\,(d/e)^d$:

$$(d!)^{1/d} \approx \left(\sqrt{2\pi d}\right)^{1/d} \cdot \frac{d}{e}.$$

As $d \to \infty$, $(\sqrt{2\pi d})^{1/d} = (2\pi d)^{1/(2d)} \to 1$, so $(d!)^{1/d} \approx d/e$. Therefore:

$$r_d \approx \frac{d}{2e}.$$

This grows linearly in d.

(c) Since $V_d^1(1) \to 0$ and $V_d^2(1) \to 0$, the unit ball becomes negligibly small in volume. To compensate and achieve a fixed volume (say 1), we must inflate the radius. The faster the volume shrinks, the faster r_d must grow.

For ℓ_∞: $V_d^\infty(r) = (2r)^d$, so $V_d^\infty(r_d) = 1$ gives $r_d = 1/2$ for all d. Since $V_d^\infty(1) \to \infty$ rather than 0, no inflation is needed — in fact the ℓ_∞-ball already has more than enough volume at any fixed radius.

Exercise 1.4 [Medium]

Let $A_d^2(r)$ denote the surface area of the ℓ_2-ball (sphere) of radius r in $\mathbb{R}^d$. It is related to the volume by

$$A_d^2(r) = \frac{d}{dr} V_d^2(r).$$

(a) Using $V_d^2(r) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} r^d$, derive a formula for $A_d^2(r)$.

(b) Compute the surface area of the unit sphere in $\mathbb{R}^3$ and $\mathbb{R}^4$.

(c) Show that $\lim_{d\to\infty} A_d^2(1) = 0$. What does this imply about the geometry of high-dimensional spheres?

Solution

(a) Differentiating $V_d^2(r) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} r^d$ with respect to r:

$$A_d^2(r) = \frac{d\,\pi^{d/2}}{\Gamma(d/2+1)} r^{d-1} = \frac{2\,\pi^{d/2}}{\Gamma(d/2)} r^{d-1},$$

where we used $\Gamma(d/2+1) = (d/2)\,\Gamma(d/2)$, so $d/\Gamma(d/2+1) = 2/\Gamma(d/2)$.

(b) For $d = 3$: $A_3^2(1) = \frac{2\pi^{3/2}}{\Gamma(3/2)} = \frac{2\pi^{3/2}}{\sqrt{\pi}/2} = 4\pi$.

For $d = 4$: $A_4^2(1) = \frac{2\pi^2}{\Gamma(2)} = \frac{2\pi^2}{1} = 2\pi^2$.

(c) Since $A_d^2(1) = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ and $\Gamma(d/2)$ grows super-exponentially (by Stirling: $\Gamma(d/2) \approx \sqrt{\pi d}\,(d/(2e))^{d/2-1/2}$), while $\pi^{d/2}$ is merely exponential, the denominator eventually dominates and $A_d^2(1) \to 0$.

This means both the volume and the surface area of the unit ℓ_2-sphere vanish in high dimensions. Despite the surface area going to zero absolutely, the *ratio* of volume near the surface to total volume goes to 1 — this is the concentration of measure phenomenon.

Exercise 1.5 [Hard]

We know $V_d^2(1) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$ and $V_d^1(1) = \frac{2^d}{d!}$.

(a) Show that for even dimensions $d = 2k$,

$$\frac{V_d^1(1)}{V_d^2(1)} = \left(\frac{2}{\sqrt{\pi}}\right)^{2k} \frac{1}{\prod_{\ell=k+1}^{2k} \ell}.$$

Hint: Express both volumes in terms of factorials and simplify.

(b) Using Stirling's formula or a direct bound, show that $\frac{V_d^1(1)}{V_d^2(1)} \to 0$ as $d \to \infty$, i.e., the ℓ_1-ball volume decays *strictly faster* than the ℓ_2-ball volume.

(c) Give a geometric interpretation: why should the ℓ_1-ball be "much smaller" than the ℓ_2-ball in high dimensions, despite the containment $B_1(0,1) \subseteq B_2(0,\sqrt{d})$?

Solution

(a) For $d = 2k$:

$$V_d^1(1) = \frac{2^{2k}}{(2k)!}, \qquad V_d^2(1) = \frac{\pi^k}{\Gamma(k+1)} = \frac{\pi^k}{k!}.$$

Taking the ratio:

$$\frac{V_d^1(1)}{V_d^2(1)} = \frac{2^{2k}}{(2k)!} \cdot \frac{k!}{\pi^k} = \frac{4^k}{\pi^k} \cdot \frac{k!}{(2k)!}.$$

Now $\frac{4^k}{\pi^k} = \left(\frac{2}{\sqrt{\pi}}\right)^{2k}$, and $\frac{k!}{(2k)!} = \frac{1}{(k+1)(k+2)\cdots(2k)} = \frac{1}{\prod_{\ell=k+1}^{2k} \ell}$.

(b) The product $\prod_{\ell=k+1}^{2k} \ell$ has k terms, each at least $k+1$, so $\prod_{\ell=k+1}^{2k} \ell \geq (k+1)^k$. Thus:

$$\frac{V_d^1(1)}{V_d^2(1)} \leq \left(\frac{4}{\pi}\right)^k \cdot \frac{1}{(k+1)^k} = \left(\frac{4}{\pi(k+1)}\right)^k \to 0 \quad \text{as } k \to \infty.$$

Since $4/\pi \approx 1.27$ is fixed while $(k+1) \to \infty$, the base of the exponential goes below 1 for $k \geq 1$, and the ratio decays exponentially.

(c) The ℓ_1-ball is the "diamond" shape: it concentrates its volume along the coordinate axes and has sharp corners along the diagonals. In high dimensions, these corners become extremely thin and contribute negligible volume. The ℓ_2-ball is rounder and distributes volume more evenly. While $B_1(0,1) \subseteq B_2(0,\sqrt{d})$, this containment uses a ball of radius $\sqrt{d}$, not the *unit* ℓ_2-ball. At unit radius, the ℓ_1-ball is strictly smaller ($B_1(0,1) \subseteq B_2(0,1)$), and the ratio of their volumes decays exponentially in d.

Concentration of Volume Under the Surface

Exercise 2.1 [Easy]

Let $B_d = B_d(0,1)$ denote the unit ℓ_2-ball in $\mathbb{R}^d$ and $B_d(0,1-\varepsilon)$ the ball of radius $1-\varepsilon$ for $0 < \varepsilon < 1$.

(a) Compute the ratio $\frac{V_d^2(1-\varepsilon)}{V_d^2(1)}$ and simplify.

(b) Evaluate this ratio for $\varepsilon = 0.05$ and $d = 10, 50, 100, 500$.

(c) In words, what happens to the fraction of volume in the thin outer shell $B_d(0,1) \setminus B_d(0,1-\varepsilon)$ as $d \to \infty$?

Solution

(a) Since $V_d^2(r) = V_d^2(1)\, r^d$:

$$\frac{V_d^2(1-\varepsilon)}{V_d^2(1)} = (1-\varepsilon)^d.$$

(b) For $\varepsilon = 0.05$:

$$d = 10: \ 0.95^{10} \approx 0.599, \quad d = 50: \ 0.95^{50} \approx 0.077,$$

$$d = 100: \ 0.95^{100} \approx 0.0059, \quad d = 500: \ 0.95^{500} \approx 7.2 \times 10^{-12}.$$

(c) As $d \to \infty$, $(1-\varepsilon)^d \to 0$ for any $\varepsilon > 0$. This means the inner ball $B_d(0,1-\varepsilon)$ contains a vanishing fraction of the total volume. Essentially *all* the volume concentrates in the thin outer shell of width ε, no matter how small ε is.

Exercise 2.2 [Medium]

Let $B_d^p(0,r)$ denote the ℓ_p-ball of radius r in $\mathbb{R}^d$.

(a) Show that for any $p \geq 1$ and any $0 < \varepsilon < 1$,

$$\frac{\mathrm{vol}\big(B_d^p(0,1-\varepsilon)\big)}{\mathrm{vol}\big(B_d^p(0,1)\big)} = (1-\varepsilon)^d.$$

Hint: Use the scaling $V_d^p(r) = V_d^p(1)\, r^d$.

(b) Fix $\varepsilon = 0.01$. Find the smallest dimension d_0 such that at least 99% of the volume of the unit ℓ_p-ball lies in the outer shell $B_d^p(0,1) \setminus B_d^p(0,1-\varepsilon)$.

(c) The result in (a) is independent of p. Give an intuitive explanation for why the concentration of volume near the surface is universal across all ℓ_p-balls.

Solution

(a) For any $p \geq 1$, $V_d^p(r) = V_d^p(1)\, r^d$, so:

$$\frac{V_d^p(1-\varepsilon)}{V_d^p(1)} = \frac{V_d^p(1)(1-\varepsilon)^d}{V_d^p(1)} = (1-\varepsilon)^d.$$

This uses only the homogeneity of the volume (scaling by r^d), not the specific form of $V_d^p(1)$.
(b) We need the fraction in the outer shell to be at least 0.99:

$$1-(1-\varepsilon)^d \geq 0.99 \iff (1-\varepsilon)^d \leq 0.01 \iff d \geq \frac{\ln 0.01}{\ln(1-\varepsilon)}.$$

For $\varepsilon = 0.01$: $\ln(0.99) \approx -0.01005$, $\ln(0.01) \approx -4.605$, so $d \geq 4.605/0.01005 \approx 458.2$, giving $d_0 = 459$.
(c) The concentration depends only on the scaling exponent d in $V_d^p(r) \propto r^d$, which is universal: the volume of *any* body in $\mathbb{R}^d$ that scales homogeneously with degree d exhibits the same shell concentration. The specific shape (p-ball, ellipsoid, etc.) determines the proportionality constant $V_d^p(1)$, but the shell ratio $(1-\varepsilon)^d$ depends only on dimensionality.

Exercise 2.3 [Hard]

Consider the unit ℓ_2-ball $B_d(0,1)$ in $\mathbb{R}^d$ and define the "equatorial slab" of thickness $\delta > 0$ as

$$S_\delta = \{x \in B_d(0,1) \mid |x_1| \leq \delta\}.$$

The volume of S_δ relative to $B_d(0,1)$ can be expressed as

$$\frac{\mathrm{vol}(S_\delta)}{\mathrm{vol}(B_d(0,1))} = \frac{\displaystyle\int_{-\delta}^{\delta} (1-t^2)^{(d-1)/2}\, dt}{\displaystyle\int_{-1}^{1} (1-t^2)^{(d-1)/2}\, dt}.$$

(a) Explain why the integrand $(1-t^2)^{(d-1)/2}$ for the cross-sectional volume appears in this formula.

(b) Show that for $|t| \leq 1$, one has $(1-t^2)^{(d-1)/2} \leq e^{-(d-1)t^2/2}$.

Hint: Use $\ln(1-u) \leq -u$ for $u \in [0,1)$.

(c) Using the bound from (b), argue that for any $\delta > 0$ fixed, the fraction of volume *outside* the slab S_δ goes to zero as $d \to \infty$. That is, almost all the volume concentrates near the equator $x_1 = 0$.

Solution

(a) For a fixed $x_1 = t$ with $|t| \leq 1$, the cross-section of $B_d(0,1)$ is the $(d-1)$-dimensional ball $\{(x_2, \ldots, x_d) : x_2^2 + \cdots + x_d^2 \leq 1 - t^2\}$, which has radius $\sqrt{1-t^2}$. Its $(d-1)$-dimensional volume is $V_{d-1}^2(1)\,(1-t^2)^{(d-1)/2}$. Integrating over t and dividing by the total volume cancels $V_{d-1}^2(1)$ (it appears in both numerator and denominator), leaving the stated ratio.
(b) We need $(1-t^2)^{(d-1)/2} \leq e^{-(d-1)t^2/2}$. Taking logarithms, this is equivalent to:

$$\frac{d-1}{2}\ln(1-t^2) \leq -\frac{(d-1)t^2}{2}.$$

Since $\ln(1-u) \le -u$ for $u \in [0,1)$ and $t^2 \in [0,1)$ for $|t| < 1$, we have $\ln(1-t^2) \le -t^2$. Multiplying by $(d-1)/2 > 0$ preserves the inequality. At $|t| = 1$ both sides are $-\infty$ (left) and $-\infty$ or 0 (right), and the bound holds trivially.
(c) The fraction of volume outside S_δ is:

$$1 - \frac{\mathrm{vol}(S_\delta)}{\mathrm{vol}(B_d)} = \frac{\int_{\delta<|t|\le 1}(1-t^2)^{(d-1)/2}\,dt}{\int_{-1}^{1}(1-t^2)^{(d-1)/2}\,dt}.$$

Using the bound from (b), the numerator satisfies:

$$\int_{\delta<|t|\le 1}(1-t^2)^{(d-1)/2}\,dt \le 2\int_{\delta}^{1} e^{-(d-1)t^2/2}\,dt \le 2e^{-(d-1)\delta^2/2}.$$

For the denominator, the central portion near $t = 0$ contributes at least:

$$\int_{-1/(d-1)^{1/2}}^{1/(d-1)^{1/2}}(1-t^2)^{(d-1)/2}\,dt \ge \frac{c}{\sqrt{d-1}}$$

for some constant $c > 0$ (from a Gaussian approximation). Thus the ratio is bounded by $C\sqrt{d}\,e^{-(d-1)\delta^2/2} \to 0$ as $d \to \infty$ for any fixed $\delta > 0$. This shows that almost all the volume is within an arbitrarily thin slab around the equator.

Stirling's Formula and Asymptotic Scaling

Exercise 3.1 [Easy]

Stirling's approximation states that for large n,

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

(a) Compute 10! exactly and via Stirling's approximation. What is the relative error?
(b) Using Stirling's formula, show that $\ln(n!) \approx n\ln n - n + \frac{1}{2}\ln(2\pi n)$ for large n. Which term dominates?

Solution

(a) Exact: $10! = 3{,}628{,}800$. Stirling: $\sqrt{20\pi}\,(10/e)^{10} = \sqrt{62.83}\cdot(3.6788)^{10} \approx 7.926 \times 4539.99 \approx 3{,}598{,}696$.
Relative error: $\frac{|3628800-3598696|}{3628800} \approx 0.0083$, about 0.83%.
(b) Taking the logarithm of Stirling's formula:

$$\ln(n!) \approx \ln\left(\sqrt{2\pi n}\right) + n\ln\left(\frac{n}{e}\right) = \frac{1}{2}\ln(2\pi n) + n\ln n - n.$$

For large n, the dominant term is $n\ln n$, since n grows faster than $\ln n$ and much faster than $\frac{1}{2}\ln(2\pi n)$. In Big-O notation: $\ln(n!) = n\ln n - n + O(\ln n)$.

Exercise 3.2 [Medium]

The volume of the unit ℓ_2-ball in $\mathbb{R}^d$ is $V_d^2(1) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$.

(a) For even $d = 2k$, we have $\Gamma(k+1) = k!$. Use Stirling's formula to show that

$$\ln V_{2k}^2(1) \approx k \ln \pi - k \ln k + k - \frac{1}{2} \ln(2\pi k)$$

for large k.

(b) Show that the dominant term in $\ln V_{2k}^2(1)$ for large k is $k \ln(\pi/k)$, which is negative for $k \geq 4$. Conclude that $V_d^2(1) \to 0$ *exponentially fast* in d.

(c) Determine the dimension d^* (approximately) at which $V_d^2(1)$ is maximized.

Hint: For even $d = 2k$, find the k where $\ln V_{2k}^2(1)$ transitions from increasing to decreasing.

Solution

(a) For $d = 2k$: $V_{2k}^2(1) = \pi^k/k!$. Using $\ln(k!) \approx k \ln k - k + \frac{1}{2} \ln(2\pi k)$:

$$\ln V_{2k}^2(1) = k \ln \pi - \ln(k!) \approx k \ln \pi - k \ln k + k - \frac{1}{2} \ln(2\pi k).$$

(b) Grouping the leading terms:

$$\ln V_{2k}^2(1) \approx k \ln\left(\frac{\pi}{k}\right) + k - \frac{1}{2} \ln(2\pi k).$$

For large k, the term $k \ln(\pi/k) = k(\ln \pi - \ln k)$ dominates. Since $\ln k > \ln \pi$ for $k \geq 4$, this is negative and grows in magnitude linearly in k. The $+k$ correction is linear but with a smaller coefficient (it just shifts the threshold), and the ln term is sublinear. Therefore $\ln V_{2k}^2(1) \to -\infty$, meaning $V_d^2(1) \to 0$ exponentially fast.

(c) The volume $V_{2k}^2(1) = \pi^k/k!$ transitions from increasing to decreasing when $\pi^{k+1}/(k+1)! < \pi^k/k!$, i.e., when $\pi/(k+1) < 1$, giving $k > \pi - 1 \approx 2.14$. So for even dimensions, the maximum is at $k = 2$ (i.e., $d = 4$) or $k = 3$ ($d = 6$). Checking: $V_4^2(1) = \pi^2/2 \approx 4.93$ and $V_6^2(1) = \pi^3/6 \approx 5.17$. Including odd dimensions, $d^* = 5$ gives $V_5^2(1) = 8\pi^2/15 \approx 5.26$. So $d^* \approx 5$.

Concentration of Gaussian Measure

Exercise 4.1 [Easy]

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$, so that $X = (X_1, \ldots, X_d)$ with independent components $X_i \sim \mathcal{N}(0, \sigma^2)$.

(a) Compute $\mathbb{E}[\|X\|^2]$ and $\mathrm{Var}(\|X\|^2)$.

Hint: $\|X\|^2 = \sum_{i=1}^d X_i^2$, each X_i^2 has mean σ^2 and variance $2\sigma^4$.

(b) Use Chebyshev's inequality to show that

$$P\big(\big|\,\|X\|^2 - d\sigma^2\big| \geq t\big) \leq \frac{2d\sigma^4}{t^2}.$$

Solution

(a) Since $\|X\|^2 = \sum_{i=1}^d X_i^2$ and the X_i are independent with $\mathbb{E}[X_i^2] = \sigma^2$ and $\mathrm{Var}(X_i^2) = 2\sigma^4$ (since $X_i^2/\sigma^2 \sim \chi_1^2$ which has variance 2):

$$\mathbb{E}[\|X\|^2] = \sum_{i=1}^d \mathbb{E}[X_i^2] = d\sigma^2, \qquad \mathrm{Var}(\|X\|^2) = \sum_{i=1}^d \mathrm{Var}(X_i^2) = 2d\sigma^4.$$

(b) Chebyshev's inequality states: for any random variable Y with finite variance, $P(|Y - \mathbb{E}[Y]| \geq t) \leq \mathrm{Var}(Y)/t^2$.
Applying this with $Y = \|X\|^2$, $\mathbb{E}[Y] = d\sigma^2$, $\mathrm{Var}(Y) = 2d\sigma^4$:

$$P\big(\big|\|X\|^2 - d\sigma^2\big| \geq t\big) \leq \frac{2d\sigma^4}{t^2}.$$

Exercise 4.2 [Medium]

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$.

(a) Using the Chebyshev bound from the previous exercise, show that

$$P\Big(\big|\,\|X\|^2 - d\sigma^2\big| \geq t\sqrt{d}\,\sigma^2\Big) \leq \frac{2}{t^2}.$$

(b) Re-interpret this as: the squared norm $\|X\|^2$ is within a factor of $1 \pm \frac{t}{\sqrt{d}}$ of $d\sigma^2$ with probability at least $1 - \frac{2}{t^2}$.

(c) For $d = 1000$ and $\sigma = 1$, find the interval $[a, b]$ such that $\|X\|^2 \in [a, b]$ with probability at least 0.95.

Solution

(a) Substituting $t \to t\sqrt{d}\,\sigma^2$ into the Chebyshev bound $P(|\|X\|^2 - d\sigma^2| \geq s) \leq 2d\sigma^4/s^2$:

$$P\Big(\big|\|X\|^2 - d\sigma^2\big| \geq t\sqrt{d}\,\sigma^2\Big) \leq \frac{2d\sigma^4}{t^2 d\sigma^4} = \frac{2}{t^2}.$$

(b) The event $|\|X\|^2 - d\sigma^2| < t\sqrt{d}\,\sigma^2$ can be rewritten as:

$$d\sigma^2\left(1 - \frac{t}{\sqrt{d}}\right) < \|X\|^2 < d\sigma^2\left(1 + \frac{t}{\sqrt{d}}\right).$$

As $d \to \infty$ for fixed t, the relative width $t/\sqrt{d} \to 0$, so $\|X\|^2$ concentrates more and more tightly around $d\sigma^2$. Equivalently, $\|X\|$ concentrates around $\sigma\sqrt{d}$.

(c) We need $2/t^2 \leq 0.05$, so $t^2 \geq 40$, i.e., $t \geq \sqrt{40} \approx 6.32$. With $d = 1000$ and $\sigma = 1$:

$$[a, b] = \Big[1000 - \sqrt{40} \cdot \sqrt{1000},\ 1000 + \sqrt{40} \cdot \sqrt{1000}\Big] = [1000 - 200, 1000 + 200] = [800, 1200].$$

So $P(\|X\|^2 \in [800, 1200]) \geq 0.95$.

Exercise 4.3 [Hard]

In machine learning, we often compute pairwise distances between data points. Consider two independent samples $X, Y \sim \mathcal{N}(0, \sigma^2 I_d)$ in $\mathbb{R}^d$.

(a) Compute $\mathbb{E}[\|X - Y\|^2]$ and $\mathrm{Var}(\|X - Y\|^2)$.

(b) Using Chebyshev's inequality, show that

$$P\left(\left|\frac{\|X - Y\|^2}{2d\sigma^2} - 1\right| \geq \varepsilon\right) \leq \frac{2}{d\varepsilon^2}.$$

(c) Similarly show that $\|X\|^2/(d\sigma^2) \to 1$ in probability. Conclude that for large d, the ratio $\|X - Y\|^2/\|X\|^2$ concentrates around a constant. What is this constant, and what does it imply for the usefulness of Euclidean distance in high-dimensional machine learning?

Solution

(a) Since X, Y are independent, $Z = X - Y \sim \mathcal{N}(0, 2\sigma^2 I_d)$. Thus:

$$\mathbb{E}[\|X - Y\|^2] = \mathbb{E}[\|Z\|^2] = d \cdot 2\sigma^2 = 2d\sigma^2.$$

$$\mathrm{Var}(\|X - Y\|^2) = \mathrm{Var}(\|Z\|^2) = 2d \cdot (2\sigma^2)^2 = 8d\sigma^4.$$

(Using the formula: for $W \sim \mathcal{N}(0, \tau^2 I_d)$, $\mathrm{Var}(\|W\|^2) = 2d\tau^4$, with $\tau^2 = 2\sigma^2$.)
(b) By Chebyshev's inequality:

$$P\big(|\|X - Y\|^2 - 2d\sigma^2| \geq s\big) \leq \frac{8d\sigma^4}{s^2}.$$

Substituting $s = \varepsilon \cdot 2d\sigma^2$:

$$P\left(\left|\frac{\|X - Y\|^2}{2d\sigma^2} - 1\right| \geq \varepsilon\right) \leq \frac{8d\sigma^4}{4d^2\sigma^4\varepsilon^2} = \frac{2}{d\varepsilon^2}.$$

(c) Similarly, $\|X\|^2/(d\sigma^2)$ satisfies:

$$P\left(\left|\frac{\|X\|^2}{d\sigma^2} - 1\right| \geq \varepsilon\right) \leq \frac{2}{d\varepsilon^2} \to 0.$$

So $\|X\|^2/(d\sigma^2) \xrightarrow{P} 1$ and $\|X - Y\|^2/(2d\sigma^2) \xrightarrow{P} 1$. Taking the ratio:

$$\frac{\|X - Y\|^2}{\|X\|^2} \xrightarrow{P} \frac{2d\sigma^2}{d\sigma^2} = 2.$$

This means that in high dimensions, the distance from any point to any other point is approximately $\sqrt{2}$ times the norm of each point. All pairwise distances become nearly equal (relative to the norms), so Euclidean distance loses its ability to distinguish "near" from "far" neighbors. This is the **curse of dimensionality** for distance-based methods like k-NN: as d grows, the contrast $\frac{d_{\max} - d_{\min}}{d_{\min}} \to 0$, making nearest-neighbor queries essentially meaningless.

Mixed Challenging Exercises

Exercise 5.1 [Hard]

Let $X \sim \mathrm{Uniform}(B_d^2(0,1))$, i.e., X is drawn uniformly from the unit ℓ_2-ball in $\mathbb{R}^d$.

(a) Using the concentration of volume near the surface, argue that $\|X\|$ is close to 1 with high probability as $d \to \infty$. Specifically, show that for any $\varepsilon > 0$,

$$P(\|X\| \leq 1 - \varepsilon) = (1 - \varepsilon)^d \to 0.$$

(b) Now consider $X \sim \mathcal{N}(0, I_d)$. From concentration results, we know $\|X\| \approx \sqrt{d}$ with high probability. If we project onto the unit sphere, $\hat{X} = X/\|X\|$, argue that the first coordinate $\hat{X}_1$ concentrates around 0 as $d \to \infty$.

Hint: $\hat{X}_1 = X_1/\|X\|$ and $X_1 \sim \mathcal{N}(0,1)$.

(c) Combine (a) and (b) with the equatorial slab result to give a coherent picture: in high dimensions, random points in the ball are near the surface, and points on the surface are near the equator (relative to any fixed direction). What does this imply about the distribution of angles between random vectors?

Solution

(a) The probability $P(\|X\| \leq 1-\varepsilon)$ equals the fraction of the ball's volume contained in $B_d^2(0, 1-\varepsilon)$:

$$P(\|X\| \leq 1-\varepsilon) = \frac{V_d^2(1-\varepsilon)}{V_d^2(1)} = (1-\varepsilon)^d \xrightarrow{d\to\infty} 0.$$

So with high probability, $\|X\| > 1-\varepsilon$. Uniform points in the ball live near the boundary.
(b) We have $\hat{X}_1 = X_1/\|X\|$ where $X_1 \sim \mathcal{N}(0,1)$ and $\|X\| \approx \sqrt{d}$ with high probability. Therefore $\hat{X}_1 \approx X_1/\sqrt{d}$.
Since X_1 is $O(1)$ (it is a standard normal), $\hat{X}_1 = O(1/\sqrt{d}) \to 0$ in probability. More precisely, for any $\varepsilon > 0$:

$$P(|\hat{X}_1| > \varepsilon) \leq P\Big(|X_1| > \varepsilon\sqrt{d}/2\Big) + P\Big(\|X\| < \sqrt{d}/2\Big).$$

Both terms go to 0 as $d \to \infty$ (the first by Gaussian tail decay, the second by concentration of $\|X\|$).
(c) The three results together paint a complete picture of high-dimensional geometry:

- (Volume near surface) Random points in the ball are effectively on the sphere.
- (Equatorial slab) Relative to any fixed axis, points on the sphere cluster around the equator ($x_1 \approx 0$).
- (Gaussian projection) Even Gaussian vectors, after normalization, have vanishing projections onto any fixed direction.

For the angle θ between two independent random vectors X, Y, we have $\cos\theta = \langle \hat{X}, \hat{Y}\rangle = \sum_{i=1}^d \hat{X}_i\hat{Y}_i$. Each term is $O(1/d)$, and the sum of d such weakly correlated terms concentrates around 0 by the law of large numbers. Therefore $\theta \approx \pi/2$ with high probability: random vectors in high dimensions are nearly orthogonal. This is another facet of the curse of dimensionality and explains why random projections (e.g., in the Johnson–Lindenstrauss lemma) approximately preserve distances.

Exercise 5.2 [Hard]

Consider a machine learning setting where you generate N data points $x_1, \ldots, x_N \sim \mathcal{N}(0, I_d)$ and classify them using a nearest-neighbor classifier.

(a) Using the result that $\|x_i - x_j\|^2/(2d) \xrightarrow{P} 1$ for all $i \neq j$, show that the relative contrast

$$\rho_d = \frac{d_{\max} - d_{\min}}{d_{\min}}, \quad \text{where } d_{\max} = \max_j \|x_1 - x_j\|,\ d_{\min} = \min_{j\neq 1} \|x_1 - x_j\|,$$

satisfies $\rho_d \xrightarrow{P} 0$ as $d \to \infty$.

(b) Suppose the data points lie on a k-dimensional linear subspace of $\mathbb{R}^d$ (with $k \ll d$). Specifically, $x_i = Az_i$ where $A \in \mathbb{R}^{d\times k}$ with orthonormal columns and $z_i \sim \mathcal{N}(0, I_k)$. Compute $\mathbb{E}[\|x_i - x_j\|^2]$ and explain why the concentration of distances now depends on k rather than d.

(c) This is the motivation behind dimensionality reduction techniques (PCA, random projections) used before applying k-NN classifiers. Explain in 2–3 sentences how the analysis from (a) and (b) justifies this preprocessing step.

Solution

(a) From the concentration result, for large d, $\|x_1 - x_j\|^2 \approx 2d$ for all $j \neq 1$, so $\|x_1 - x_j\| \approx$

$\sqrt{2d}$. More precisely, for any $\varepsilon > 0$ and fixed N:

$$P\left(\max_j \left|\frac{\|x_1 - x_j\|^2}{2d} - 1\right| > \varepsilon\right) \leq \sum_{j=2}^{N} \frac{2}{d\varepsilon^2} = \frac{2(N-1)}{d\varepsilon^2} \to 0.$$

So with high probability, all distances lie in $[\sqrt{2d(1-\varepsilon)}, \sqrt{2d(1+\varepsilon)}]$, giving:

$$\rho_d \leq \frac{\sqrt{1+\varepsilon} - \sqrt{1-\varepsilon}}{\sqrt{1-\varepsilon}} \approx \frac{\varepsilon}{\sqrt{1-\varepsilon}} \to 0$$

as $\varepsilon \to 0$ (which we can take since $d \to \infty$). Hence $\rho_d \xrightarrow{P} 0$.

(b) Since A has orthonormal columns, $\|x_i - x_j\|^2 = \|A(z_i - z_j)\|^2 = \|z_i - z_j\|^2$ (orthonormal matrices preserve norms). With $z_i - z_j \sim \mathcal{N}(0, 2I_k)$:

$$\mathbb{E}[\|x_i - x_j\|^2] = \mathbb{E}[\|z_i - z_j\|^2] = 2k.$$

The concentration bound becomes $P(|\|x_i - x_j\|^2/(2k) - 1| > \varepsilon) \leq 2/(k\varepsilon^2)$. The effective dimension for distance concentration is k, not d. If k is moderate (say $k = 50$), there is meaningful contrast between distances even if $d = 10{,}000$.

(c) In the ambient space $\mathbb{R}^d$, distance-based classifiers like k-NN fail because all pairwise distances become nearly equal (the contrast $\rho_d \to 0$). If the data actually lives on or near a low-dimensional subspace, projecting to that subspace recovers the true intrinsic distances, where concentration is governed by $k \ll d$ and meaningful distance contrast is preserved. This is why PCA or random projections before k-NN can dramatically improve classification performance.